Rudra Patel

Cybersecurity × Machine Learning · Full-Stack Developer

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SUMMARY

B.Tech Computer Science & Engineering student at CSPIT, CHARUSAT focused on where machine learning meets cybersecurity — phishing and anomaly detection, secure coding and threat modeling. I build and deploy full-stack products end to end: three of my projects are live.

EDUCATION

B.Tech, Computer Science & Engineering — CSPIT, CHARUSAT — 2024 – 2028 (expected) · currently Semester 5

PROJECTS

KarnaKavach — AI/ML phishing email & malicious URL detection — karna-kavach.vercel.app
React · TypeScript · Node.js/Express · Python · scikit-learn · Gemini API · Firebase

- Fused a 20-rule detection engine, a TF-IDF (1–3-gram) + Logistic Regression model and Gemini reasoning into one 0–100 risk score (25/35/40 weighting, automatic fallback to rules + ML when AI is unavailable).
- Built a feedback-driven retraining pipeline with model versioning and logged metrics; classifies 8 threat categories with explainable reasons and PDF reports.
- Secured user data with Firebase Auth (email, Google, GitHub) and owner-only Firestore security rules.

Aryam Maps — full-stack e-commerce store for handcrafted wooden maps — aryam-creation.vercel.app
Next.js 16 · React 19 · TypeScript · Supabase (Postgres + Storage) · Razorpay · Tailwind CSS

- 28 pages and 24 API routes: live product customiser, cart, checkout and order tracking, plus a 13-screen admin console.
- Integrated Razorpay with server-side HMAC-SHA256 signature verification and webhooks; added TOTP two-factor login for admins.
- Stale-while-revalidate product cache, in-browser WebP image compression and printable A4 business reports (jsPDF).

KrishiIntel — agricultural market intelligence (SerpApi hackathon, team of 2) — krishi-intel-cody.onrender.com
Next.js · Express · TypeScript · Prisma · MySQL · Socket.IO · SerpApi · JWT

- Ranks mandis by net realisation (freight, handling, cess, spoilage) and a weighted opportunity score; farmer, FPO and admin portals with server-side RBAC.
- Integrated SerpApi Google Search, News and Maps with an in-memory TTL cache, provenance badges and real-time Socket.IO shipment tracking.

Also built: VendorBridge (procurement ERP with RBAC and Supabase RLS), GlobeTrotter (Gemini-powered multi-city travel planner).

SKILLS

Languages	C++, Java, Python, JavaScript, TypeScript, Bash
Security	Linux, Wireshark, Burp Suite, Nmap, Metasploit; secure coding, threat modeling
ML & Data	scikit-learn, NumPy, pandas, Jupyter, Gemini API
Web	React, Next.js, Node.js, Express, Tailwind CSS
Data & Cloud	Supabase, Firebase, MySQL, Prisma, Vercel, Render, Docker, Git

CURRENTLY LEARNING

C++ (STL, memory model) · DSA/DAA with regular LeetCode practice · vulnerability classes and exploitation · ML for anomaly detection · secure coding.